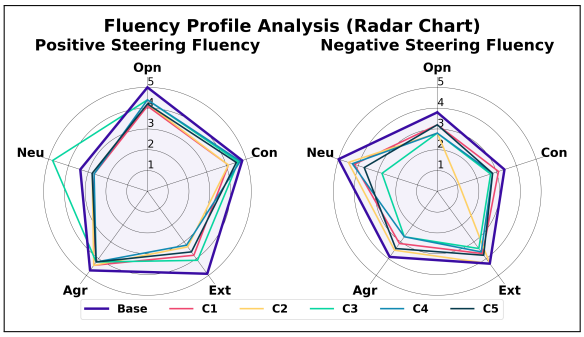

Figure 1: **Fluency profile analysis for conditions C0-C5** compared against Base steering. *Fluency degradation for both Positive and Negative steering can be observed across all traits for all the Conditional Methods used*. Although trait shifts that were comparable to the base values, the significant degradation of fluency suggests the need to use orthogonalised vectors carefully for steering purpose.

tions base(C0), C4 and C5. Additionally, we analyze *fluency* and *variance* scores under identical measurement conditions, using LLaMA as the reference model.

Steering strength remains largely conserved under progressive orthonormalisation (C1–C5)(Table 1). However, Figure 1 demonstrates how fluency scores consistently degrade as geometric constraints are enforced. Comparing to baseline fluency values reported in (Bhandari et al., 2026) (High/Low), Openness drops from $5.0/3.8$ to $4.1/3.2$ (C1), $4.3/2.8$ (C2), and remains around $4.4/2.8$ through C3–C5. Similarly, conscientiousness decreases from $4.8/3.5$ to $4.3/3.1$ in C1 following identical trends to Openness for other conditions, while Extraversion exhibits the largest decline, from $4.9/4.3$ to $3.8/3.6$.

These results indicate that although orthonormalisation preserves directional steering magnitude, it removes shared components necessary for fluent and expressive generation, leading to reduced variance and degraded output quality. Hence, to explain our **RQ2**: we conclude that explicitly enforcing geometric independence does not improve steering effectiveness and instead introduces a quality–control trade-off.

4.2 RQ3: Cross-Model Consistency of Trait Dependencies

To assess whether observed trait dependencies generalise across model families, we compare steering behaviour between *LLaMA-3-8B-Instruct* and *Mistral-8B-Instruct* under identical conditions (C0, C4, C5) using the same extraction, constraint, and evaluation pipeline. Across both models, we observe consistent geometric patterns: several traits exhibit substantial cross-trait bleed even after enforcing geometric constraints. For example, **Openness** shows high maximum bleed values in both models, with $B_{\max}$ remaining large under hard orthonormalisation (C5), e.g., ≈ 3.0 in LLaMA-3 and ≈ 2.8 in Mistral-8B. Similarly, **Extraversion** and **Agreeableness** continue to induce strong off-diagonal effects across conditions, indicating that these dependencies are not artifacts of a single model but reflect shared structure in personality representations.

At the same time, we observe clear model-specific modulation in steering responsiveness. Most notably, **Neuroticism** exhibits strong target steering in LLaMA-3 ($T \approx 3.1$–3.2 across C0–C5), whereas Mistral-8B shows low target response ($T \approx 0.0$–0.7) under the same conditions, despite comparable geometric treatment. Importantly, this suppression persists even when geometric independence is enforced (C5), suggesting that the absence of behavioural response cannot be attributed solely to vector entanglement. Together, these results indicate that while cross-trait dependencies are largely consistent across model families, their behavioural expression is shaped by model-specific training and alignment constraints rather than geometry alone. A detailed table for all the observations (C1-C5) is provided in Appendix A.

5 Conclusion

Given the popularity of steering methods in the literature, we systematically analysed the behaviour of steering vectors under various conditions. Our method considered the Big Five Personality traits, and we investigated whether personality steering directions in large language models are geometrically independent, and how enforcing geometric constraints affects the steering behaviours. Through the analysis across two model families (Llama and Mistral), we show that personality traits are not independent directions in activation space. Even when strong global constraints such as symmetric orthonormalisation are applied, steering one trait consistently induces measurable changes in other unintended traits, indicating persistent cross-trait dependencies.

Limitations

This work studies personality steering behaviour using a limited set of large language models and personality datasets, and future work could extend the analysis to a broader range of model families and trait representations. While we focus on Big Five traits and judge-based evaluation, additional datasets and alternative evaluation frameworks may reveal further structure in trait interactions. Our analysis relies on linear geometric constraints; exploring other orthogonalisation or projection methods could provide a more comprehensive understanding of trait disentanglement. Finally, we use LLMs as judges for behavioural assessment, and incorporating human evaluation or complementary metrics is left for future investigation.

6 Ethical Considerations

Steering large language models using latent vectors introduces ethical considerations, particularly when such steering is applied in uncontrolled or unsupervised settings. Steering vectors are learned approximations of complex behavioural traits and do not provide transparent or complete representations of the values they encode; as a result, unintended attributes or hidden information may be co-activated during steering. This raises concerns about value misalignment, especially in real-world deployments where subtle behavioural shifts could have downstream social or psychological impacts. Additionally, aggressive or poorly understood steering may bypass safety mechanisms or distort model behaviour in ways that are difficult to detect or reverse. These risks highlight the importance of careful evaluation, interpretability, and constraint-aware steering when modifying model behaviour.

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